Traffic signal recognition using YOLO, a comparative analysis

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Abstract- **The identification of traffic signs is a major challenge for intelligent automobiles. Recognition of traffic signs gives useful data, such as alerts and directions, for cooperative intelligent transport systems (CITS) and advanced driver assistance systems (ADAS). In realistic autonomous driving scenarios, traffic signs can be challenging to identify even with a very precise real-time method. As a result, drivers are calling for more secure and trustworthy traffic sign recognition technology, and current automakers are also working on developing one. All images captured by the actual driving car are blurry and distorted. The study of traffic sign recognition systems faces several challenges due to external, unmanageable components, such as the impact of unfavorable conditions, and the practical use is still in its infancy. This study provides the YOLO model for traffic sign identification as a solution for the issues mentioned above. The first step is to generally arrange the traffic signs into various categories. Next, the prepossessing process then follows the characteristics of the different kinds of signs. The improved convolution neural network receives the altered images and breaks down the categories into more specific subcategories. The YOLOv3-tiny and YOLOv4-tiny algorithms are examined by using the CCTSDB dataset and the Tsinghua-Tecent 100k(TTK100) dataset and YOLOv2,YOLOv3 andYOLOv5 are examined using the CCTSDB and HRRSD(High Resolution Remote Sensing Detection) datasets depending on the various categories. The findings indicate that YOLOv4-tiny algorithm achieves good recognition performance on both TTK100 and CCTSDB datasets and YOLOv5 on both CCTSDB and HRRSD dataset are most suited for use in traffic sign recognition systems and has a high classification accuracy.**

Keywords—*YOLO, Detection, Segmentation, Recognition, FPS.*

I. Introduction

The rapid advancement of science and technology along with the economy, cars have progressively become a part of most households because of the fast expansion of the car sector, which offers both positive and negative impacts.

A key element of the smart vehicle driving system is traffic sign recognition and it is one of the most important areas of computer vision research. The development rate of automobiles is significantly faster than the rate of road construction, traffic congestion and safety hazards on the roads are getting worse. Typically, exercises involving the recognition of traffic signs are carried out in settings that are naturally occurring; nevertheless, inclement weather, such as rain, snow, or fog, can make it difficult to read the information on the signs, and overexposure and low light typically make them less visible. Therefore, the quick development of intelligent transportation systems offers strong technological support for issues of this nature. It successfully integrates science and technology, which include, information technology, data transmission technology, sensor technology, and AI used in transportation, service control, and car manufacturing. Governments, professionals, and academics both domestically and internationally are paying increasing attention to and using this integrated transportation system, which is now an internationally acknowledged essential method to solve general traffic problems.

Early traffic sign identification techniques traversed the full image and produced numerous candidate locations using a sliding window strategy. The candidate locations were then found using a variety of predetermined features, including SIFT (scale-invariant features) [3], LBP (local binary pattern) [4], and HOGs (histograms of oriented gradients) [2].Then, for detection and identification, these features were uo to here into a powerful classifier, like an Adaboost [6], a Random Forest [7], or an SVM (support vector machine) [5]. Now,the detection architecture are targeted by employing a lightweight convolutional neural network, the detection time for traffic sign recognition in deep convolutional neural network is decreased. Different YOLO models work faster at target detection than the majority of the current, complex deep convolutional neural network architectures.

Therefore, employing various data sets applied to the various Yolo models, this study compares the various Yolo-based models with the deep convolutional neural network-based traffic sign identification approaches on the basis of their accuracy. The remaining portions of the manuscript are arranged as follows: A quick overview of the literature is provided in Section II regarding the models in the YOLO family and how they are used in the area of object detection for traffic sign detection; Section III presents the Model Performance Evaluation which has different data types, field study of them, and a correlation; and Finally, some concluding remarks are presented in Section IV.

II. Literature Survey

One of the examples of a one-stage target detection technique is the YOLO series algorithm. There have been eight primary versions proposed since the first generation YOLOv1 algorithm was first proposed in 2015, following recent years of research. An end-to-end neural network is suggested by the You Only Look Once (YOLO) project in order to simultaneously forecast bounding boxes and probability by class. It differs from past object identification systems' approach, which employed classifiers as detectors. The same model has been offered in multiple fresh iterations, each of which builds upon and enhances the previous one. This

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timeline in "Fig. 1" shows how YOLO has developed in recent years.

A. YOLO

The YOLO CNN is a form of convolutional neural network that enables the identification and classification of objects in the shape of bounding boxes. A bounding box is the smallest rectangle that will fit the fully recognized item. On the single-shot theory, YOLO operates. In other words, all objects are identified simultaneously in one pass of the image because of the network architecture. "Fig. 2" depicts the layout of YOLO CNN. [9] YOLO input is given a three-channel image that has been scaled down to 448x448 pixels. The image is initially converted by passing it through a section of the altered GoogleNet architecture. Following conversion, we are left with feature maps that are 14x14x1024. Another convolution is then carried out. The outcome is applied twice in a completely connected layer, changing to a 1470x1 dimension and becoming a tensor of 7x7x30. A resulting detection is achieved when the obtained tensor has been placed through a detection method. The tensor in the image is a 7x7 mesh display. A cell's 40 values contain the following information: 20 values depict the relationship to each of the 20 classes that are available, whereas 10 values define two potential frames. All of this data has been filtered, and the results are shown.

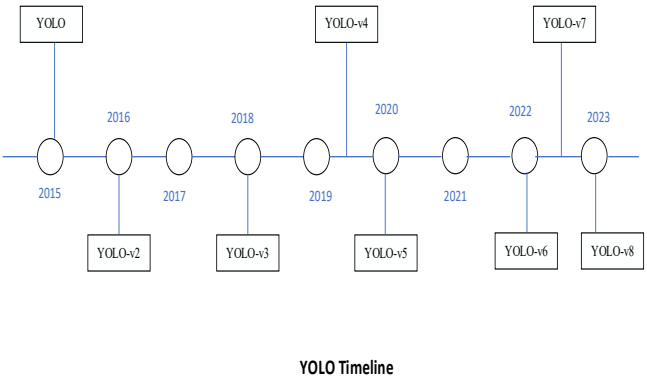

Fig.1 Figure depicts a significant development in Object detection with YOLO since 2015.

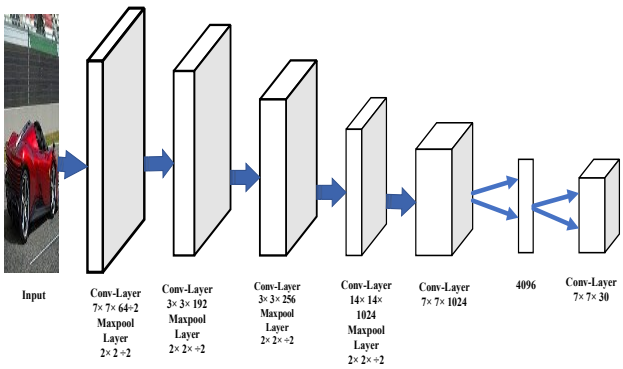

Fig.2. Architecture of YOLO CNN

B. YOLOv2

It was released in 2016. This is YOLO's second version, which combines a variety of design decisions from past works with brand-new concepts to improve the efficiency and precision of YOLO. YOLOv2 [10] eliminated Yolo1's limitations and More efficient than Faster R in real-time applications in both CNN and SDD(Single-Shot Detector). It supports instruction on multiple scales. But less precise than its competitors' SSD and RetinaNet.

C. YOLOv3

It was released in 2018. To overcome the limitations of earlier models, it is an upgraded version of YOLO and YOLO2.It is more likely to find small objects, make predictions on several scales, and have greater efficiency than SDD. [11] Yolov3 surpassed earlier models by adding connections between the layers of the backbone network, introducing an objectness measures into bounding box prediction, and producing predictions at three distinct levels of information.

D. YOLOv4

It came with updates like better feature aggregation, a bag of freebies, mish activation, and a bag of goodies when it was released in April 2020. As options for aggregating features, YOLOv4 considers the neck using FPN (Feature Pyramid Network), PAN (Pipeline Assisted Neural Network), NAS-FPN (Neural Architecture Search FPN), BiFPN (Bi-directional FPN), ASFF (Adaptively Spatial Feature Fusion), and SFAM (Simplified Fuzzy Adaptive Resonance Theory Map). Therefore, the fundamental structure of YOLOv4 [12] has 53 convolutional layers.

E. YOLOv4-tiny

By reducing model parameters to speed up identification and streamlining the network architecture, [12]YOLOv4-tiny satisfies the requirements for deployment on mobile or embedded devices. The backbone network continues to use the CSPBlock (Cross Stage Partial Block) module, and For feature extraction, the YOLOv4 small approach uses the CSPDarknet53-tiny network.

F. YOLOv5

YOLOv5 is the first member of the YOLO family to be created without a paper companion [13], which was released in June 2020. It is also still undergoing development on its repo. As someone develop their model, the YOLOv5 models train quickly, which decreases the cost of experimentation. You can quickly transform YOLOv5 for Pytorch weights to ONXX weights(Open standard for machine weight) to CoreML to ios and infer on single photos, batches of images, video feeds, or camera ports.

G. YOLOv6

In YOLOv6, extra layers in the network that are placed between these features and the final head have been

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experimentally shown to improve performance. The YOLOv6 repository also includes architectural costs and several training pipeline enhancements, including anchor-free training, SimOTA tag assignment, and SIoU box regression loss.

H. YOLOv7

It becomes available in November 2022. By taking into account the amount of memory necessary to keep layers in memory as well as the amount of time it takes a gradient to back propagate through the layers, the authors of YOLOv7 improve on earlier studies in this area. Depending on the gradient, their network will be able to learn more efficiently. They chose E-ELAN, an extended version of the ELAN (Efficient Long Range Attention Network) computational block, as their final layer aggregate.YOLOv7-tiny, YOLOv7, and YOLOv7-W6 are the three primary models that comprise the YOLOv7 [14] architecture and each of which has been designed to work in a particular execution environment (either an edge GPU, a standard GPU(Graphics Processing Unit), or a cloud GPU).

I. YOLOv8

YOLOv8 [15] was launched in the month of January 2023 by Ultralytics, the company that developed YOLOv5. There were five different sized versions available: YOLOv8n (nano), YOLOv8s (small), YOLOv8m (medium), YOLOv8l (large), and YOLOv8x (extra enormous). YOLOv8 covers a variety of visual tasks, including object recognition, segmentation, posture estimation, tracking, and classification. Additionally, it gave rise to the significant and revolutionary YOLOv5 model. A user-friendly CLI (Command Line Interface) and a well organized Python package are only two of the architectural and developer experience modifications and enhancements made to YOLOv8 from YOLOv5. As an anchor-free model, YOLOv8 predicts an object's center as opposed to its offset from a predetermined anchor box.

III. Model Performance Evaluation

A. Experimental data set (TT100K, CCTSDB)

The testing dataset was obtained from two datasets: One was CCTSDB (CSUST Chinese Traffic Sign Detection Benchmark) dataset and second was Tsinghua-Tencent 100k Tutorial (TT100k) [8], that was created by a Tsinghua University research team. The TT100K dataset contains 45 different types of traffic signs that were manually annotated on intercepted and photographed scenes of street in various lighting and environmental circumstances. The validation set has 3073 photos, whereas the training set has 6107 images. The three scale feature photos' final sizes are 52 x 52, 26 x 26, and 13 x 13, respectively. The input image is cropped to 416 x 416, but the picture quality in TT100K is 2024 x 2024.The another dataset, which is divided into a training set with 15724 images and a testing set with 400 images, has three distinct types of signs: warning, prohibitory, and required. To conduct the ablation experiments, we used the data from the CCTSDB.

B. Comparative Analysis of YOLOv4-tint model and YOLOv3-tiny model by using TTK100 dataset.

The precision of identification and detection speed of YOLOv4 are faster and more accurate than YOLOv3-tiny belongs to the category of lightweight network types that are similar to it, with mAP rising by 2.4% and FPS reaching 58 f/s. The model size has also been reduced to 23.8MB for simpler installation on mobile or embedded devices.The findings signify that the enhanced method outperforms YOLOv3-tiny in a number of performance metrics Table 1 experiments further choose the most popular two-stage detection network using the TTK100 dataset and compare the experimental methods for YOLOv3 and YOLOv4.

Table 1: The TTK100 dataset is used to compare various experimental algorithms.

Algorithm	mAP(%)	FPS(f/s)	Model Size(MB)
YOLOv3-tiny	81.4	41	34.9
YOLOv4-tiny[12]	83.8	58	23.8

C. Comparative analysis of YOLOv3-tint model and YOLOv4-tiny model using the CCTSDB dataset

The YOLOv4-tiny has the maximum frame rates (FPS) when compared to the YOLOv3-tiny, but model memory usage increases to 21.7MB and the map is also shrunk by 10.17%. The model with the highest results for memory capacity was the YOLOv3-tiny.

Table 2: Results of comparison experiments on the CCTSDB dataset

Models	mAP(%)	FPS(f/s)	Model Size(MB)
YOLOv3-tiny	74.40	219	21.7MB
YOLOv4-tiny[12]	84.57	32.73	6.6MB

D. Experimental data set (CCTSDB)

The CCTSDB [16] collection, which contains data on traffic signs in China, was collected there. “Fig. 3” illustrates its three prohibition, warning, and instruction types. To assure the correctness of the results, a significant quantity of data must be used to aid the dataset. There are 16,123 photos in CCTSDB data collection; in which 10,123 sets are used for training, 3,000 sets for verification , and 3,000 sets for testing. They all were captured in actual environments, which is more consistent with the traffic sign recognition in today's complicated situations.

E. Comparative Analysis of YOLOv3,YOLOv4 and YOLOv5 using CCTSDB dataset.

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When the same kind of lightweight network's detection speed and identification accuracy are measured against the CCTSDB dataset. The YOLOv3 one-stage algorithm showed favorable results for detection speed and accuracy. On CCTSDB, the detection precision is 96% and the FPS is 73 f/s.As shown in Table 3, the most recent algorithms, the YOLOv4 and YOLOv5, have great detection accuracies and processing speeds when identifying traffic signs.

Fig 3. Sample of the CCTSDB dataset.

Table 3. Comparison results on different algorithms by using CCTSDB dataset.

Model	P%	R%	mAP@0.5%	FPS(f/s)
YOLOv3	88.1	94.6	96	73
YOLOv4	88.1	92.8	95.8	78
YOLOv5	84.9	95.2	95.4	85

F. Experimental data Set (HRRSD)

The Chinese Academy of Sciences Xian Institute of Optics and Fine Mechanics made available the HRRSD (High Resolution Remote Sensing Detection) small target open dataset. Each target fully satisfies the requirements of a tiny target, making up between 35 and 80 pixels of the entire dataset. Ship, track and field, airplane, bridge, vehicle, intersection, and another categories make up the majority of the 13 categories in the HRRSD small target dataset. As seen in Table 4, each category is split up into a complete verification set training set, and a testing set.

Table 4. HRRSD dataset details

S.No	Name	Training set	Val set	Test set
1	Ship	950	948	1988
2	Bridge	1123	1121	2326
3	Ground track field	859	856	2017
4	Storage tank	1099	1092	2215
5	Basketball court	923	920	2233
6	Tennis court	1043	1040	2212
7	airplane	1226	1222	2451
8	Baseball diamond	1007	1004	2022
9	Harbor	967	964	1953
10	Vehicle	1188	1186	2382
11	Crossroad	903	901	2219
12	T junction	1066	1065	2289
13	Parking lot	1241	1237	2480

G. Comparative Analysis of YOLOv2 and YOLOv3 using the CCTSDB dataset

When the various models on the HRRSD dataset are compared. The detection accuracy of the YOLOv2 [10] is 79.2% with an FPS of 80 f/s, whereas the detection accuracy of the YOLOv3 [11] is higher at 81.2% with an FPS of 110 f/s, which is shown in Table 5.

Table 5. HRRSD dataset comparison result on models.

Model	mAP@0.5%	FPS (f/s)
YOLOv2	79.2	80
YOLOv3	81.2	110

IV. *Conclusion*

In this study, different Yolo algorithms are analyzed for traffic signal recognition. Furthermore, this work compared and analyzed the different YOLO algorithms. In this comparative study, we observed that YOLOv4-tint is more easily installed on mobile or embedded devices due to its higher recognition accuracy, 83.8% mAP, 58 f/s FPS, and model size of 23.8 MB on the TTK100 dataset. and 219 frames per second FPS with a memory size of 21.7 MB, 74.40% mAP, and CCTSDB dataset. We have also compared YOLOv2, YOLOv3, and YOLOv5 on the CCTSDB dataset, and found that YOLOv5 had good identification accuracy with 84.9 P%, 95.2 R%, 95.4 mAP%, and 85 f/s FPS, and YOLOv3 had 81.2% mAP, 110f/s FPS on HRRSD dataset. Because there are still differences amongst YOLO approaches, in order to improve algorithm performance, the next stage of research for traffic sign identification will focus on striking a balance between detection speed and identification accuracy.

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